

面向 On-the-fly NVS 的逐帧生命周期建模与风险感知接纳框架

Abstract

现有新视角合成与 3D Gaussian Splatting 方法通常依赖离线相机标定或全局优化流程，难以满足大规模场景的在线重建需求。近期 **On-the-fly NVS** 系统通过**轻量级位姿初始化、直接高斯基元采样、联合位姿—表示优化以及 anchor-based 增量场景管理**，实现了从**无位姿输入有序图像到可渲染 3DGS 表示**的在线构建。然而，我们观察到，在**长序列、开放场景及大旋转等困难数据集**上，该类系统的失败并不总是表现为单个测试视角的低质量渲染，而是更早地发生在输入流的接纳边界、位姿初始化链条以及几何—表示状态传播过程中。传统基于平均 PSNR、SSIM 和 LPIPS 的评估方式难以刻画这种在线系统特有的**时序失稳行为**。

本文提出一种**面向在线新视角合成的逐帧状态风险评估与自适应接纳框架**。首先，我们构建统一的输入流生命周期主表，将**每个输入帧在系统中的完整生命周期显式记录下来**，包括**关键帧接纳、位姿初始化、几何支撑、局部优化稳定性、表示支撑以及测试窗口质量**等信息。在此基础上，我们在线失效建模为一种状态传播风险，而非孤立的坏帧检测问题，并设计**逐帧风险评估器**，从到帧即得的**运动与匹配信息、位姿初始化后的 PnP / mini-BA 诊断量，以及历史窗口中的时序退化趋势**中估计当前帧的接纳风险。进一步地，我们引入风险感知的**三态接纳策略**：对稳定且有表示价值的帧立即接纳，对当前几何支撑不足但未来可能恢复的帧延迟保留，对低价值或高风险帧直接丢弃。被延迟的候选帧不会立即扰动主状态链，而是以局部几何假设和表示缺口信息的形式保留，并在后续邻近帧成功注册、局部表示增强或 anchor 上下文成熟时，通过轻量级 pose-first recovery 重新尝试并入系统。

该框架将 On-the-fly NVS 的失败分析从传统的结果级质量评估推进到输入流级生命周期建模，能够区分**后期退化、渐进恶化和局部 episode 式失稳**等不同失败形式。与原始硬接纳机制相比，本文方法旨在降低错误位姿对在线状态链的持续污染，提升长序列和困难场景下的几何稳定性与渲染鲁棒性。本文为在线增量式 NVS 系统提供了一种新的风险感知建模范式，即不再仅追求平均重建质量，而是显式优化输入流中每一帧对未来几何—表示状态传播的影响。

1. Introduction

动机

Baseline:

《On-the-fly Reconstruction for Large-Scale Novel View Synthesis from Unposed Images》

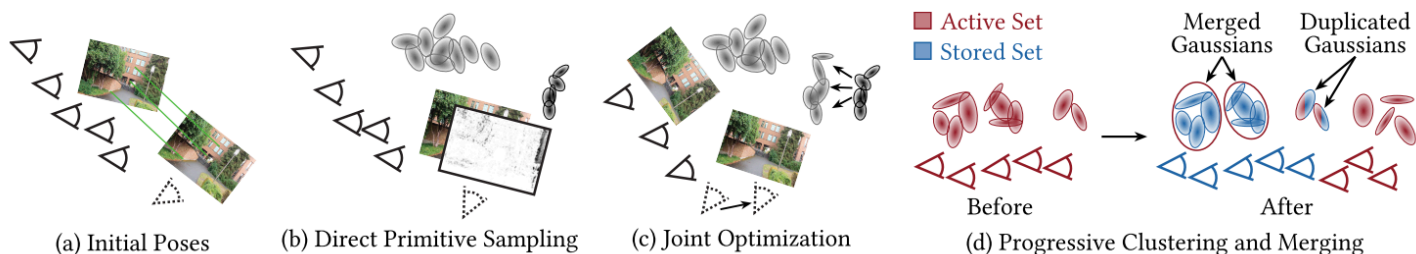


Fig. 2. Overview of our method. As each new image arrives, (a) we first use learning-based feature matching, reformulated in a GPU-friendly manner to achieve fast initial pose estimation (b) we perform direct sampling of Gaussian primitives by estimating the probability that a pixel should generate a Gaussian, and also sample the size, (c) the two first steps enable fast and effective joint optimization, improving both poses and scene representation and (d) we incrementally cluster into *anchors*, and merge nodes allowing us to represent large-scale scenes. When placing an anchor, the Gaussians from the active set are stored for later rendering. We create the new active set by merging primitives that appear small and duplicating the others, reducing the number of Gaussians being optimized.

传统问题

只看**最终平均质量**：**PSNR**、**SSIM**、**LPIPS**，关键数据被这种隐式表达掩盖掉，无法突出重建核心问题

Table 1. Reconstruction time and novel view quality results for different methods. The first section compares our method with others that use unposed images, while the second section employs Structure from Motion (SfM). Reported runtimes include pose optimization. COLMAP intrinsics are used for both Photo-SLAM and MonoGS. The **best** and **second best** are color coded for pose-free methods.

	TUM				MipNeRF360				STATICHIKES			
	PSNR↑	SSIM↑	LPIPS↓	Time↓	PSNR↑	SSIM↑	LPIPS↓	Time↓	PSNR↑	SSIM↑	LPIPS↓	Time↓
Photo-SLAM	19.30	0.700	0.382	0:02:12	16.54	0.505	0.603	0:02:11	14.13	0.316	0.660	0:02:01
MonoGS	16.60	0.682	0.381	0:16:18	14.46	0.436	0.663	0:04:05	15.46	0.301	0.659	0:09:19
Ours	23.02	0.821	0.250	0:00:50	24.31	0.775	0.300	0:01:02	20.40	0.589	0.365	0:01:30
GLoMAP + Taming 3DGS (7k)	25.29	0.868	0.191	0:03:33	27.52	0.866	0.226	0:08:50	20.23	0.537	0.465	1:02:34
COLMAP + 3DGS (7k)	24.75	0.874	0.175	0:04:11	27.82	0.877	0.212	0:09:52	20.40	0.538	0.463	2:33:57
COLMAP + 3DGS (30k)	25.34	0.881	0.157	0:09:44	29.66	0.906	0.170	0:26:05	24.08	0.757	0.267	2:47:11

Table 2. Novel view quality results for different methods that require low-resolution input. We use COLMAP intrinsics for COLMAP Free 3DGS.

	TUM				MipNeRF360				STATICHIKES			
	PSNR↑	SSIM↑	LPIPS↓	Time↓	PSNR↑	SSIM↑	LPIPS↓	Time↓	PSNR↑	SSIM↑	LPIPS↓	Time↓
DROID-Splat	19.49	0.721	0.325	0:06:35	25.87	0.776	0.258	0:11:18	19.65	0.470	0.506	0:09:26
CF-3DGS	15.05	0.578	0.405	1:10:27	13.52	0.295	0.621	1:08:14	15.21	0.301	0.560	7:52:54
Ours	22.45	0.815	0.225	0:01:11	25.80	0.834	0.182	0:00:55	21.93	0.673	0.270	0:01:32

baseline的评估方式

PSNR

衡量重建图像和真值图像的像素误差大小，先把重建图和真值图逐像素做差，计算整体的**平均平方误差（MSE）**，再把这个误差换算成对数尺度的**“信噪比”**来表示，数值越高通常表示越接近真值

SSIM

衡量两张图在结构/亮度/对比度上的相似度，更关注**“结构是否看起来一样”**，在局部窗口里分别比较两张图的**亮度（均值）、对比度（方差）、结构（协方差/相关性）**，把这三项综合成一个相似度分数再对全图取平均，结果在0~1之间，数值越高越好

LPIPS

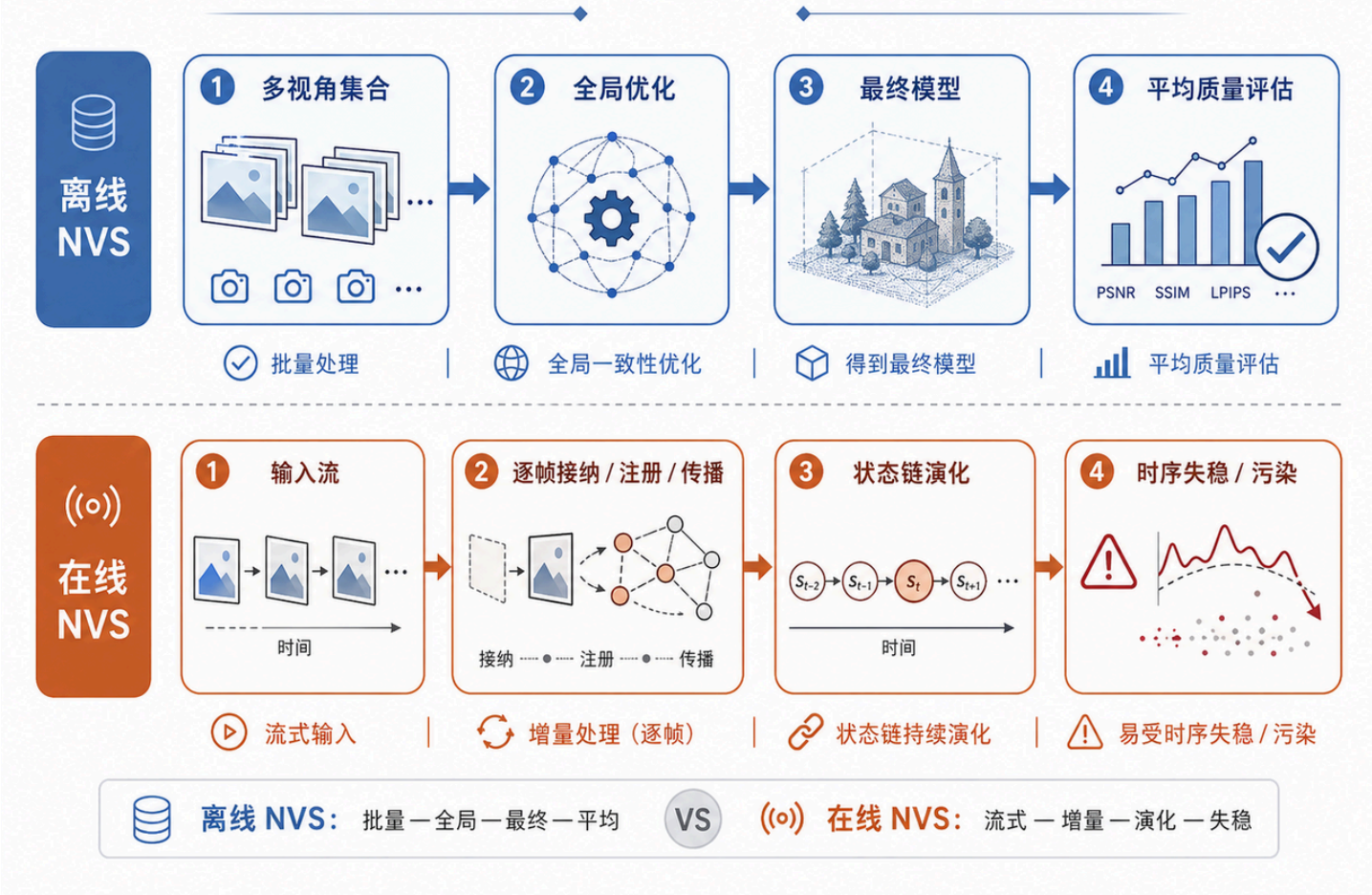
用深度网络特征衡量**“人眼感知上的差异”**，把两张图分别输入一个预训练的深度网络（常用VGG/AlexNet等），在多个层取特征图，计算**特征差异并做加权汇总**，得到符合人类感知的**距离分数**，结果在0~1之间，数值越低表示视觉上越像真值

在线增量系统

输入的每一帧图像“命运”不同，可能：

- 被直接过滤掉
- 进入位姿初始化
- 注册失败
- 注册成功并进入状态链
- 对后续几何和表示产生长期影响
- ...

离线 NVS 与 在线 NVS 的核心差异

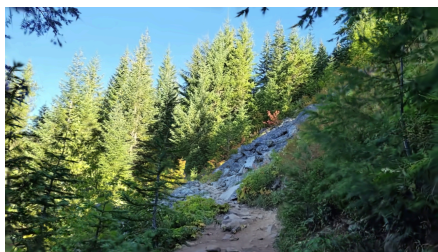


结果级质量评估适合离线方法，但不足以解释在线系统的时序失稳
需要把『失败对象』从“坏结果”提升为“输入流中的状态传播风险”

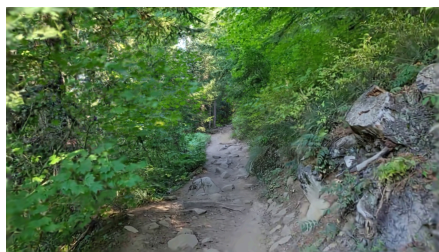
数据集的差异性

在对测试帧 (test images) 的逐帧指标整理后，发现困难数据集 (hard datasets) 上的失效并不是同一种模式

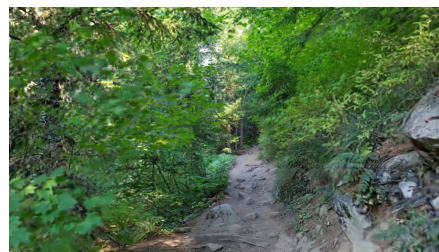
- **forest1**: 不是一开始就差，而是到了后半段明显更差，属于**长时序后期退化**；



第11帧



第471帧



第481帧

- university2: 不是突然崩，而是整体逐步走差，属于**渐进式恶化**；



第561帧



第711帧

- TUM_long: 不是全程差，而是在局部区间出现明显异常，属于**局部间断式失稳**



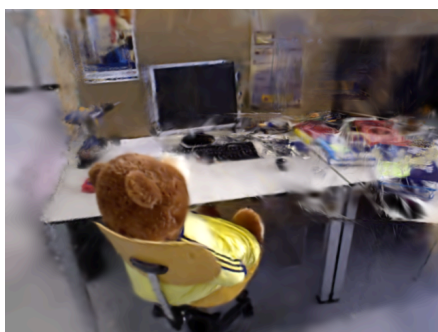
第481帧



第601帧



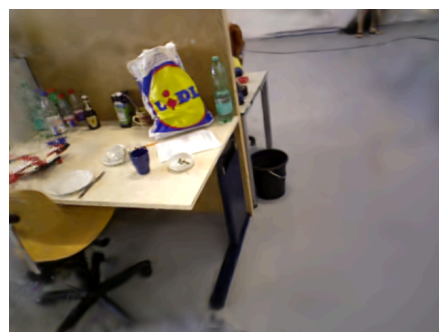
第631帧



第1261帧



第1771帧



第1801帧

✿ 对于在线增量重建，真正需要被刻画的不只是平均质量，而是渲染失败在时间维度上的呈现方式，并且输入帧进入系统后对后续几何-表示状态传播是否造成持续扰动

核心目标

2. Related work

State-Space Machine (SSM状态机)

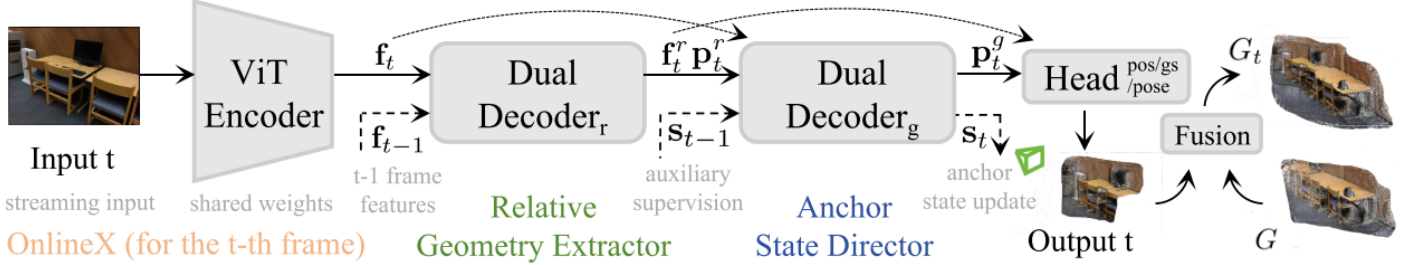


Figure 2. **Overall architecture of OnlineX.** Our framework features a two-stage, active-to-stable pipeline. First, the Relative Geometry Extractor processes consecutive frames to capture high-fidelity active relative information. The Anchor State Director then uses this local information to recurrently update its stable global state, yielding a globally consistent representation for the final output. The diagram illustrates this process for a single time step, which would be sequentially repeated for each frame in the input stream. Dashed lines represent information passed from the previous time step or carried over to the next.

OnlineX

State Update

Substituting the routing mask $\mathbf{M}_t$ into Eq. (8):

$$\mathbf{S}_t = \mathbf{M}_t \odot \hat{\mathbf{S}}_t + (1 - \mathbf{M}_t) \odot \mathbf{S}_{t-1} \quad (12)$$

Tokens within the selected patches are fully replaced by the decoder output; all others are exactly preserved. This is the binary-gate instance of Eq. (8) discussed in Sec. 3.2.

State Update with Test-Time Training

Routing mask $\mathbf{M}_t$ determines where to write, yet within the selected patches every token is still fully overwritten. We can further modulate how much to write by combining $\mathbf{M}_t$ with the attention-derived learning rate β_t from Eq. (4):

$$\mathbf{S}_t = (\mathbf{M}_t \odot \beta_t) \odot \hat{\mathbf{S}}_t + (1 - \mathbf{M}_t \odot \beta_t) \odot \mathbf{S}_{t-1} \quad (13)$$

For unselected patches $\mathbf{M}_t = 0$, the state is exactly preserved regardless of β_t ; for selected patches, β_t provides token-level soft modulation. This combination is still training-free since β_t is derived from the existing decoder's cross-attention weights. Note that $\mathbf{M}_t \odot \beta_t \in [0, 1]$, so Eq. (13) remains equivalent to the unified gate framework in Eq. (8).

MeMix

Algorithm. MeMix Inference

Require: Input sequence $\{I_t\}_{t=1}^T$, base model f , patch partition $\{P_j\}_{j=1}^p$

- 1: $\mathbf{S} \leftarrow \mathbf{S}_0$
- 2: $\mathcal{M} \leftarrow \emptyset$
- 3: **for** $t = 1$ to T **do**
- 4: $\mathbf{X}_t \leftarrow \text{Tokenize}(I_t)$
- 5: $\hat{\mathbf{S}}_t, \mathbf{Y}_t \leftarrow \text{cross attn}(\mathbf{S}_{t-1}, \mathbf{X}_t)$
- 6: $r_t \leftarrow \text{RouteScore}(\hat{\mathbf{S}}_t, \mathbf{X}_t)$
- 7: $\mathcal{P}_t \leftarrow \text{Bottom-K}(r_t)$
- 8: $\mathbf{M}_t \leftarrow \text{Gate}(\mathcal{P}_t)$
- 9: $\mathbf{S}_t \leftarrow \mathbf{M}_t \odot \hat{\mathbf{S}}_t + (1 - \mathbf{M}_t) \odot \mathbf{S}_{t-1}$
- 10: $\mathcal{M}_t \leftarrow \text{Head}(\mathbf{Y}_t)$
- 11: **end for**
- 12: **return** $\mathcal{M}$

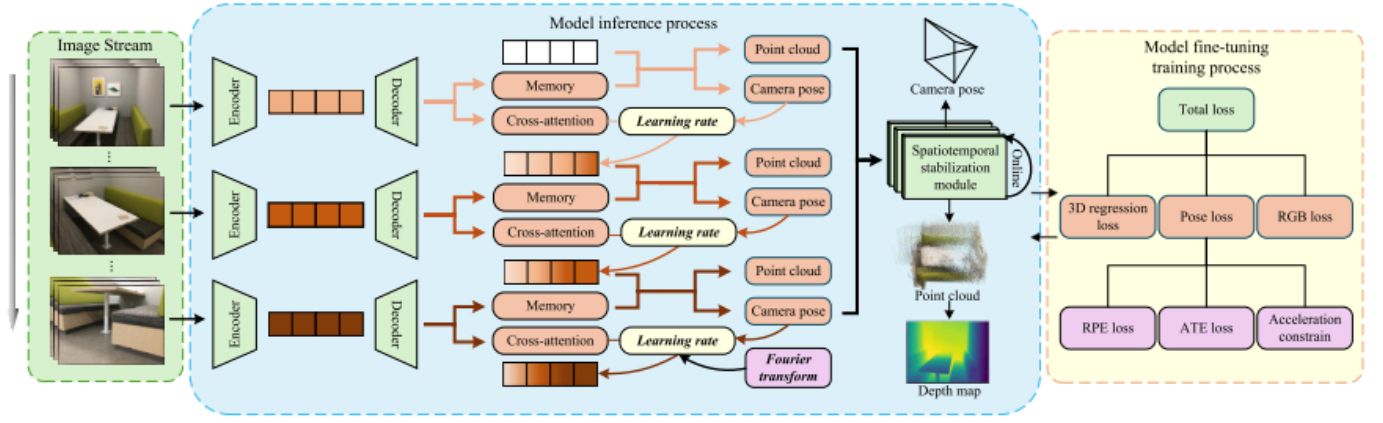


Fig. 1: The overall pipeline. Given a streaming image sequence, the model maintains a persistent memory that is iteratively updated through cross-attention with incoming frames. PAS3R introduces pose-adaptive state update modulation, where the update intensity (learning rate) is dynamically adjusted based on camera motion and image structure (via Fourier analysis), allowing the model to balance adaptation to novel viewpoints and preservation of accumulated geometry. During inference, the model predicts camera poses and dense point clouds at each step. During training, we further enforce trajectory-consistent optimization using a novel pose loss with ATE, RPE, and acceleration constraints. A lightweight online spatiotemporal stabilization module is applied to reduce trajectory jitter and geometric artifacts. Together, these components enable stable and scalable long-horizon streaming 3D reconstruction.

PAS3R

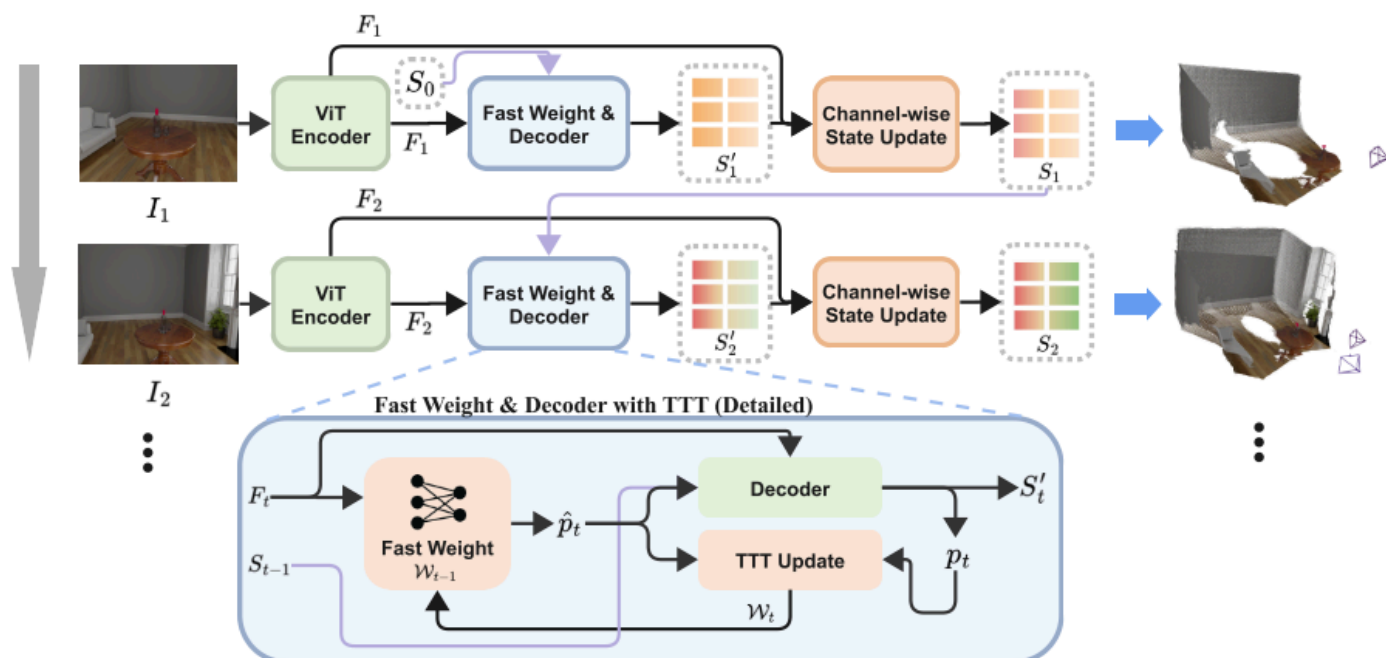


Figure 2: Overview of Mem3R. **Top:** For each frame I_t in the streaming image sequence, a ViT encoder extracts image features F_t . The MLP-based fast-weight module $\mathcal{W}$ performs camera tracking with decoder under Test-Time Training (TTT), while the fixed-size state $\mathcal{S}$ and decoder preserve and update geometric information, producing an intermediate state S'_t . S'_t is then fused with the previous state S_{t-1} through a channel-wise update module (see Sec. 3.3) to obtain the final updated state S_t . By replacing pose-related state token with $\mathcal{W}$, Mem3R maintains a constant-size recurrent state while reducing long-term temporal forgetting in streaming 3D reconstruction. **Bottom:** Illustration of the fast-weight and decoder module with TTT (see Sec. 3.2).

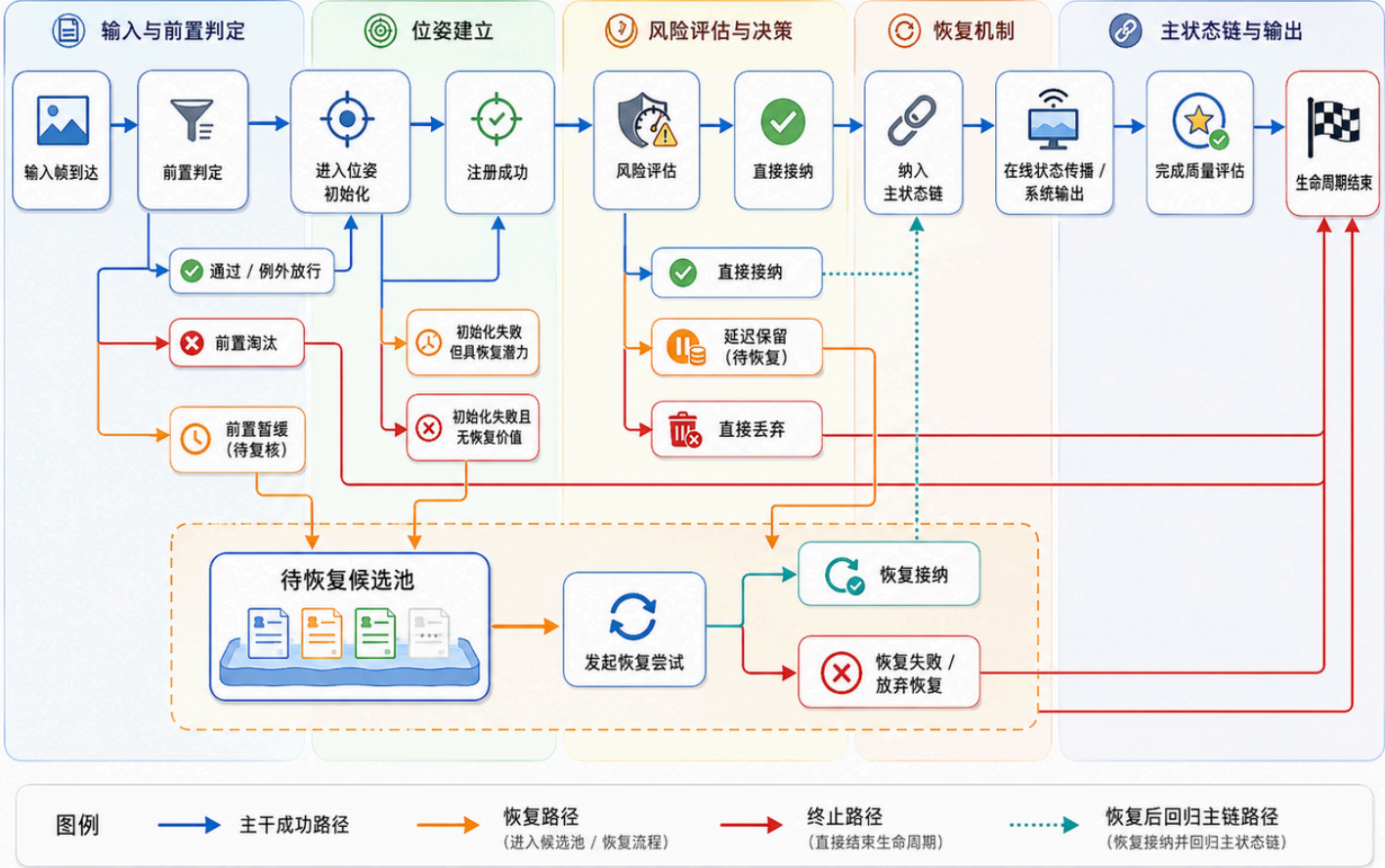
Mem3R

含义

- 输入当前所在的位置 (当前状态)
- 下一步可以去哪里 (未来可能的状态)
- 以及哪些变化会将当前输入带到下一个状态 (发生了什么)

3. Method

在线逐帧生命周期状态机主干框架图



本文不把『失败对象』看作孤立坏帧，而把它视为输入流中可传播的状态风险
因此，方法设计遵循“先显式建模生命周期，再评估风险，最后进行风险感知接纳与恢复”的统一链条

持续更新重建

引入一个逐帧生命周期状态机，用于把在线系统中每个输入帧从到达退出的处理过程显式记录下来
特点：

- 每一帧都会有一个状态机
- 状态机的表达是显示的（包含身份信息与生命周期状态）

状态机做“语义表达”，主表记录“完整事实”

显式状态与隐式诊断量的本质区别

显式状态：表达的是生命周期身份

隐式诊断量：表达的是状态形成的依据、强弱和原因

字段组	典型内容
帧身份	frame_id, arrival_order
前置判定	gate result, match count, displacement
位姿建立	pose-init attempt, registration
几何诊断	correspondences, inlier ratio, reprojection error
表示支撑	active support, anchor context
测试窗口	PSNR, SSIM, LPIPS, pose error

聚合中间量设计阈值：

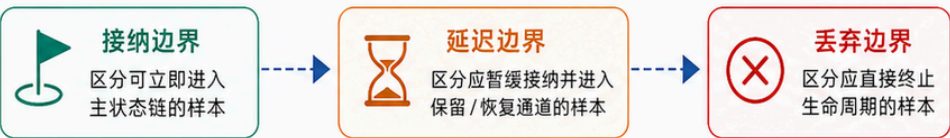
- 风险得分 ↓ 接纳代价
- 价值得分 ↑ 接纳收益
- 可恢复性得分 ↑ 延迟潜力

核心得分与决策逻辑

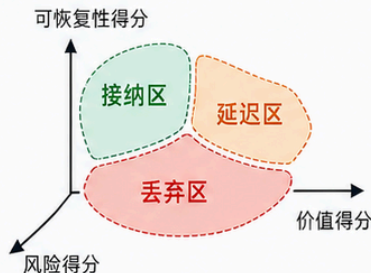
将分散的隐式诊断证据压缩为三个可决策量，用于支持直接接纳、延迟保留和直接丢弃。



4 阈值设计目标 (Threshold Design Goals)



目标：围绕三类关键分界，确定三个核心得分的判定区间。



每个已成功注册的输入帧 I_t 定义：

决策向量

$[R_t, V_t, Q_t]$

- R_t ：主导接纳边界
- V_t ：约束收益
- Q_t ：只用于判定延迟恢复

基础变量：接纳风险得分 R_t （当前立即接纳的污染风险）

$R_t = \phi_R(\mathcal{D}_t)$

其中：

$\mathcal{D}_t$ ：当前帧的风险相关隐式诊断证据集合

- 位姿/几何不确定性
- 状态链支撑不足
- 历史时序退化

R_t 高：立即接纳更可能污染后续状态链

R_t 低：当前接纳相对安全

基础变量：接纳价值得分 V_t （当前帧进入系统后带来的边际收益）

$V_t = \phi_V(\mathcal{D}_t)$

其中：

$\mathcal{D}_t$ ：当前帧的价值相关隐式诊断证据集合

- 表示补充
- 视角/运动互补
- 链上支撑增益

V_t 高：该帧进入系统有明显收益

V_t 低：当前帧边际贡献有限

派生变量：可恢复性得分 Q_t （当前不接纳但保留等待恢复是否值得）

$Q_t = \psi(B(R_t), V_t, C_t)$

其中：

$B(R_t)$ ："风险处于可恢复区间"的程度

C_t ：当前可观测的恢复上下文条件

R_t 低：直接接纳；

R_t 高：不可救；

V_t 低：不值得保留；

C_t 高：未来上下文确实可能补救

Q_t 高：当前不接纳也值得保留等待后续恢复

Q_t 低：延迟保留意义不大

🌸 三类得分均被约束在 $[0, 1]$ 范围内

$$a_t = \begin{cases} \text{直接接纳,} & R_t \leq \tau_R^{\text{low}}, V_t \geq \tau_V, \\ \text{延迟保留,} & B(R_t) \geq \tau_B, V_t \geq \tau_V^{\text{min}}, Q_t \geq \tau_Q, \\ \text{直接丢弃,} & \text{otherwise.} \end{cases}$$

4. Experiment

1. 诊断性实验

1.1 证明 baseline 重建质量低下主要由位姿误差导致，而非纯高斯表示问题

2. 上限性试验

2.1 通过更强pose参考，估计“只改位姿”能带来的上限

3. 方法有效性试验

3.1 比较：

- baseline 硬接纳
- 风险感知接纳
- 风险感知接纳 + 恢复

看长序列 hard scenes 上是否减少污染和恶化

4. 消融实验

4.1 Ablation

- 去掉生命周期建模
- 去掉风险分数某一项
- 去掉恢复机制
- 去掉延迟保留机制

看各自贡献

5. 评价指标：

- PSNR , SSIM , LPIPS
- rel_trans_error , rel_rot_error_deg
- abs_trans_error , abs_rot_error_deg

Table 3. **Camera pose estimation on ScanNet.** We report Absolute Translation Error (ATE), Relative Translation Error (RPE trans), and Relative Rotation Error (RPE rot).

Method	ATE↓	RPE trans↓	RPE rot ↓
Spann3R [34]	<u>0.096</u>	0.023	0.661
CUT3R [35]	0.099	<u>0.022</u>	<u>0.600</u>
Ours	0.085	0.019	0.550

Table 3: **Ablations on routing policy and score design.** The Default row corresponds to Bottom-k + Dot + score($\hat{\mathbf{S}}_t, \mathbf{X}_t$); other rows change one component at a time.

Variant	Depth@KITTI		Camera@TUM			3R@NRGBD		
	AbsRel ↓	$\delta < 1.25$ ↑	ATE ↓	RPE _{trans} ↓	RPE _{rot} ↓	Acc ↓	Comp ↓	NC ↑
Default (TTT3R with MeMix)	0.103	92.1	0.028	0.013	0.376	0.099	0.020	0.616
Patch selection								
Top-k	0.102	91.8	0.068	0.021	0.595	0.178	0.049	0.587
Random-k	0.108	91.4	0.028	0.013	0.382	0.102	0.023	0.616
Scoring function								
Cosine ($S_t^{\cos}$)	0.105	91.6	0.028	0.013	0.375	0.099	0.023	0.616
Attn (S_t^{attn})	0.107	90.7	0.030	0.014	0.418	0.105	0.025	0.611
Update strategy								
Full-update	0.108	91.4	0.035	0.015	0.443	0.127	0.034	0.604
No-update	0.114	89.0	0.162	0.066	1.624	0.461	0.672	0.528
Routing score								
score($\mathbf{S}_{t-1}, \mathbf{X}_t$)	0.107	91.1	0.039	0.016	0.437	0.111	0.027	0.608
score($\mathbf{S}_{t-1}, \mathbf{Y}_t$)	0.124	85.9	0.041	0.016	0.426	0.249	0.055	0.575
score($\hat{\mathbf{S}}_t, \mathbf{Y}_t$)	0.128	84.7	0.046	0.046	0.465	0.308	0.112	0.573

MeMix

Table 1: Quantitative comparison on Sintel [7] and TUM [54] datasets. **Bold** indicates the best performance, and underlined indicates the second best. ↓ indicates lower is better.

Method	Sintel			Tum		
	ATE ↓	RPE trans ↓	RPE rot ↓	ATE ↓	RPE trans ↓	RPE rot ↓
Mem4D [10]	0.263	0.091	0.812	0.061	0.020	0.517
CUT3R [69]	0.209	<u>0.071</u>	0.632	0.047	0.015	0.448
TTT3R [13]	<u>0.210</u>	0.090	0.730	<u>0.029</u>	0.013	<u>0.379</u>
IVGGT [81]	0.237	0.096	0.806	0.027	<u>0.012</u>	0.313
Ours	0.211	0.053	<u>0.688</u>	0.027	0.011	0.475

PAS3R

Table 2: Quantitative comparison on Sintel, Bonn [39], and Kitti [23] datasets. **Bold** indicates the best performance, and underlined indicates the second best. ↓ indicates lower is better, while ↑ indicates higher is better.

Method	Sintel		Bonn		Kitti	
	Abs Rel ↓	$\sigma < 1.25$ ↑	Abs Rel ↓	$\sigma < 1.25$ ↑	Abs Rel ↓	$\sigma < 1.25$ ↑
Mem4D [10]	0.520	43.1	0.072	95.7	0.140	82.0
CUT3R [69]	0.432	47.0	0.077	93.8	0.122	87.6
TTT3R [13]	<u>0.406</u>	48.8	0.069	95.4	0.114	<u>90.6</u>
IVGGT [81]	0.329	66.0	0.059	97.4	0.185	69.8
Ours	0.407	48.7	<u>0.064</u>	<u>96.6</u>	<u>0.115</u>	91.4

PAS3R

Table 1: Quantitative evaluation of video depth estimation on long sequences from the KITTI dataset. **Bold** indicates the best result per column within each depth category, and green indicates improved or maintained performance of our model over its respective base counterpart.

Method	300 frames		350 frames		400 frames		450 frames		500 frames	
	Abs Rel ↓	$\delta_{1.25}$ ↑	Abs Rel ↓	$\delta_{1.25}$ ↑	Abs Rel ↓	$\delta_{1.25}$ ↑	Abs Rel ↓	$\delta_{1.25}$ ↑	Abs Rel ↓	$\delta_{1.25}$ ↑
Metric Depth										
CUT3R	0.13	83.8	0.14	82.7	0.14	82.1	0.15	81.1	0.15	80.4
Ours	0.12	88.4	0.12	87.4	0.13	87.1	0.13	86.8	0.13	86.0
TTT3R	0.12	88.5	0.12	87.7	0.13	87.3	0.13	86.6	0.13	86.5
Ours + TTT3R	0.11	89.5	0.12	88.9	0.12	88.6	0.12	88.2	0.13	87.5
TTSA3R	0.12	88.8	0.12	88.0	0.13	87.5	0.13	86.7	0.13	86.5
Ours + TTSA3R	0.12	89.0	0.12	88.5	0.12	88.2	0.13	87.7	0.13	87.0
Scale-Invariant Depth										
CUT3R	0.12	87.4	0.13	87.0	0.13	87.1	0.13	86.7	0.13	86.5
Ours	0.11	90.3	0.11	89.6	0.11	89.4	0.11	89.2	0.11	89.1
TTT3R	0.11	90.2	0.11	89.6	0.12	89.5	0.12	89.1	0.12	89.1
Ours + TTT3R	0.10	91.3	0.11	90.8	0.11	90.5	0.11	90.1	0.11	89.9
TTSA3R	0.11	90.9	0.11	90.2	0.11	90.0	0.12	89.5	0.12	89.4
Ours + TTSA3R	0.10	92.0	0.10	91.5	0.11	91.2	0.11	90.7	0.11	90.5

Mem3R

- δ (预测深度和真实深度之间，相对误差落在 1.25 阈值以内的像素占比) ?

$$\max \left(\frac{d_{gt}}{d_{est}}, \frac{d_{est}}{d_{gt}} \right) < 1.25 \quad \uparrow$$

6. 代价分析

增加的计算开销、延迟和存储成本是多少

5. Conclusion